Vasopressors, Inotropes, Push Dose Medications & Paralytics

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Learning Objectives

- Describe the pharmacology and clinical indications of vasoactive medications
- Explain the role of inotropes in managing shock and cardiac dysfunction
- Describe the rationale, indication and preparation of push-dose medications
- Discuss the mechanism and appropriate use of neuromuscular blocking agents
- Select the appropriate IV fluids for trauma, shock and critical illness.

Vasopressors

Medications that cause targeted vasoconstriction of the blood vessels increasing systemic vascular resistance (SVR) and improve mean arterial pressure (MAP)

They are used in states of shock or severe hypotension when fluid resuscitation is not enough to maintain adequate perfusion of vital organs

Most commonly effective in the following shock states where hypotension is caused by vasodilation or impaired vascular tone in distributive shock states such as:

Septic

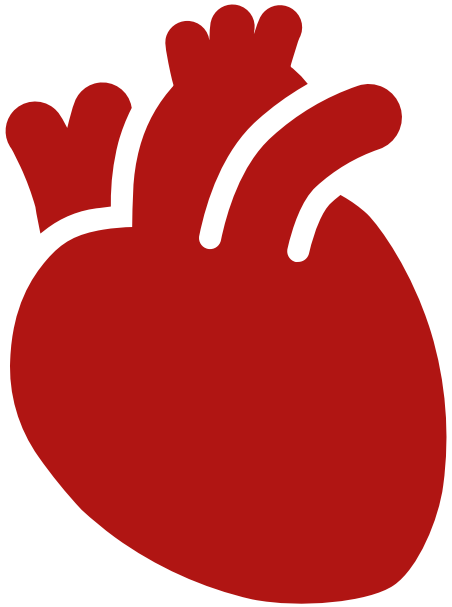
Anaphylactic

Neurogenic

The are also used in cardiogenic or obstructive shock when MAP is critically low helping to maintain organ perfusion until definitive treatment can be obtained.

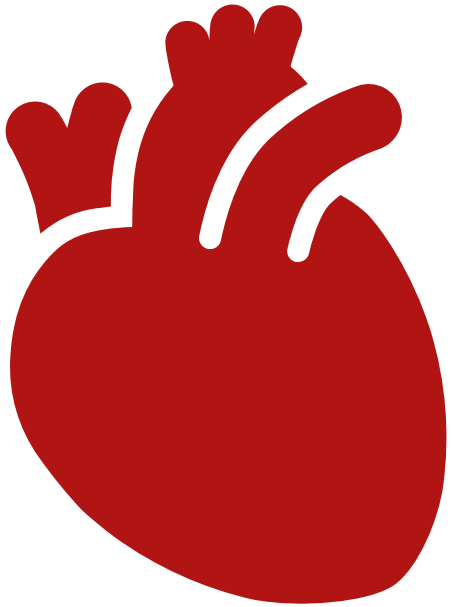
How do they work?

- Vasopressors act on the adrenergic receptors in the cardiovascular system:
  - **Alpha1 Receptors- "Squeeze the Pipes"**
    - Cause peripheral vasoconstriction
    - Increase SVR & MAP
  - **Beta1 Receptors- "Boost the Pump"**
    - Increase heartrate and contractility
    - Improve cardiac output



How do they work?

- **Beta 2 Receptors-** "Relax some vascular beds"
  - Cause vasodilation (particularly skeletal muscle vasculature
  - Smaller role in most vasopressor effects, but relevant in mixed action medications like epinephrine.
- **V1 receptors (vasopressin)** "non-adrenergic" vasoconstriction
  - Constrict smooth muscle through different pathways (non adrenergic)
  - Useful when catecholamines (i.e. Norepinephrine) are less effective.



Common Vasopressors

Norepinephrine (Levophed)

- First-line in most shock states
- Strong A1, and some B1 response
- Used primarily in sepsis
- **8-12 mcg/min titrate to get adequate response**
- **2-4mcg/min for maintenance**

Epinephrine (Adrenaline)

- Strong A1, B1 & B2
- Used in anaphylaxis, cardiac arrest and refractory shock
- Easy to administer in preloaded syringes and as a rescue pressor.
- **2-10 mcg/min or**
- **10-20mcg IVP as a rescue medication**

Common Vasopressors

Phenylephrine

- Pure A1
- Good for vasodilatory hypotension when tachycardia is an issue
- **10-35 mcg/min as infusion or**
- **40-200 mcg as a push dose MD ONLY**
- **0.5 -1.4 mcg/kg/min**

Vasopressin

- V1 receptor mediated vasoconstriction
- Added on in septic shock
- Was recently used in ACLS but has been since removed in 2020
- **0.02-0.04 units/min (12-24 units/hr)**

Inotropes

Inotropes are medications that change the strength (contractility) of the heart's contraction.

In emergency and critical-care we use positive inotropes to improve cardiac output when the heart is failing to pump effectively.

Positive inotropes work by increasing intracellular calcium availability in cardiac muscle which strengthens each heartbeat and improves blood flow.

This works primarily through

B1 adrenergic stimulation –increase contractility and sometimes HR	Phosphodiesterase-3 inhibition-increasing cyclic AMP resulting in stronger contractions	Direct effects on cardiac monocytes (digitalis compounds)

What do they do?

- Increase cardiac output (CO)
- Improve tissue perfusion when CO is low
- Support circulation in cardiogenic shock, decompensated heart failure or post cardiac arrest
- Can increase myocardial oxygen demand * ***these must be used with caution****

Most Common Positive Inotropes

Dobutamine

- Strong B1 agonist
- Increases cardiac contractility with mild vasodilation
- Great for cardiogenic shock or acute heart failure
- **2-10 mcg/kg/min may go up to 20mcg/kg/min**

Epinephrine

- B1 & B2 and A1 effects
- Increases cardiac contractility, HR and vasoconstriction
- Used in cardiac arrest, anaphylaxis and refractory shock
- **08-12 mcg/min**

Norepinephrine

- Primarily a vasopressor but has B1 inotropic effects
- Improves CO when hypotension limits perfusion
- **2-10 mcg/min**

Most Common Positive Inotropes

Dopamine

- Dose dependant effects
  - Moderate doses B1 (inotropic)
  - High does A1 (pressor)
- Historically used now less common due to arrhythmias
- Easy to use as it comes premixed in many cases (adult and pediatric dosing available)
- **2-10 mcg/kg/min titrate to effect up to 20mcg/kg/min**

Milrinone

- PDE-3 inhibitor
- Inotrope + vasodilator (Inodilator)
- Used primarily in heart failure, RV failure or post cardiac surgery
- **Loading does 50mcg/kg over 10 min then infusion 0.125-0.75 mcg/kg/min**

Digoxin

- Increases intracellular calcium through Na+/K+ -ATPase inhibition
- Slower onset, more chronic use
- Helps in heart failure and rate control for AF
- **IV loading dose 8-12 mcg/kg**
- **typical adult dose is 0.5-1mg**

Push Dose Pressors

What are push dose pressors and why do we use them?

Rapid correction of sudden hypotension

- Useful when BP drops unexpectedly because of RSI, sedation or peri-intubation

Bridge to infusion pressor

- Buys time in an urgent situation if the continuous infusion is not prepared yet or delivered

Treat transient, predictable hypotension

- Ideal for procedural sedation (i.e. propofol) where transient vasodilation is expected

Immediate availability in emergencies

- Can be mixed directly from crash cart medications when no vasopressor infusion is immediately accessible

Short-acting, titratable effect

- Fast onset (approx.-1 min) and short duration (5-20 min) makes them well-suited for stabilizing rapidly changing hemodynamics

Push-Dose Epi from a Vial (2 Steps)

9mL Saline

1mg/mL Epi

STEP 1: Pull 1mL of 1mg/mL Epi (Vial) & Mix with 9cc Saline =

100mcg/mL

1cc Epi

9mL Saline

STEP 2: Mix 1cc of (100mcg/mL) Epi with 9cc Saline =

10mcg/mL

EPINEPHRINE

Has alpha and beta1/2 effects so it is an inopressor
Do not give cardiac arrest doses (1 mg) to patients with a pulse

Mixing Instructions:

- Take a 10 ml syringe with 9 ml of normal saline
- Into this syringe, draw up 1 ml of epinephrine from the cardiac amp (Cardiac amp contains Epinephrine 100 mcg/ml)
- Now you have 10 mls of Epinephrine 10 mcg/ml

Onset-1 minute
Duration-5-10 minutes
Dose-0.5-2 ml every 2-5 minutes (5-20 mcg)

PHENYLEPHRINE

It is pure alpha, so no intrinsic inotropy, and *no increase in heart rate*, but increases in coronary perfusion can improve cardiac output.

Mixing Instructions:

- Take a 3 ml syringe and draw up 1 ml of phenylephrine from the vial (vial contains phenylephrine 10 mg/ml)
- Inject this into a 100 ml bag of NS
- Now you have 100 mls of phenylephrine 100 mcg/ml
- Draw up some into a syringe; each ml in the syringe is 100 mcg/ml

Onset-1 minute
Duration- 10-20 minutes
Dose-0.5-2 ml every 2-5 minutes (50-200 mcg)

	Phenylephrine	Epinephrine
Target push-dose syringe concentration	*100 mcg/mL (cc)* *(0.5-2 cc pushes)*	*10 mcg/mL (cc)* *(0.5-2 cc pushes)*
Mechanism	Alpha agonist	Alpha & B 1/2 agonist
Dose	50-200 mcg (q 2-5 min)	5-20 mcg (q 2-5 min)
Onset	1 min	1 min
Duration	10-20 min	5-10 min
Indication	Hypotension *with* tachycardia	Hypotension *without* tachycardia
Notable uses	• Refractory A-fib with RVR • Septic shock • Neurogenic shock • Hemorrhagic shock	• Cardiogenic shock • Cardiac arrest with ROSC • Anaphylactic shock • Hemorrhagic shock • Septic shock • Neurogenic shock

IV push medications

Rapid onset of action

- Delivers immediate medication into the blood stream producing an immediate therapeutic effect

Precise titrated dosing

- Allows clinicians to give small controlled amounts of medication and assess response in real time.
- Ideal when the patient's hemodynamics are rapidly changing or when a smaller does is required.

Useful in time critical situations-Cardiac arrest, RSI

Bridge to infusions

- Push-dose medications provide temporary stabilization while awaiting infusion preparation
- Helpful when no infusion pump is readily available
- The clinical situation is brief or the medication is short acting.

IV Push Dose Medications

- How to deliver?
  - Diluted or undiluted ?
  - Slow or fast?
- Compatible/Not compatible
- Labelling
- Types
  - Vasopressors
  - Paralytics
  - Pain
  - Sedation
  - Antibiotics

BC WOMEN'S HOSPITAL+ HEALTH CENTRE

Providence HEALTH CARE — How you want to be treated.

Vancouver CoastalHealth

Y-SITE INJECTION DRUG COMPATIBILITY CHART for CRITICAL CARE wards

KEY TO SYMBOLS:

I = **Incompatible; DO NOT MIX**

C = **Compatible;** may be mixed via Y-site. Compatibility depends upon many factors including temperature, pH, IV fluid, concentration, order of mixing and brand of drug. ***Check tubing below Y-site carefully for discoloration, cloudiness or precipitation***

☐ = (Blank) **DO NOT MIX;** conflicting or no compatibility information available

✱ = **Call Pharmacy**; compatibility depends on concentration

	amiodarone	ampicillin	calcium chloride	calcium gluconate	ceFAZolin	ceftTAZIDime	ceftriAXONE	ceFURoxime	ciprofloxacin	clindamycin	cycloSPORINE	dexmedeTOMidine 4 mcg/mL	DOBUTamine	DOPamine	EPInephrine	famotidine	fentanyl	fluCONazole	furosemide	gentamicin	heparin	hydralazine	hydrocortisone sod succ	HYDROmorphone	insulin regular human	ketamine	labetalol	lidocaine	magnesium sulfate	meropenem	metronidazole	midazolam	milrinone	morphine	nitroglycerin	nitroprusside	NORepinephrine	penicillin G Sodium	piperacillin-tazobactam	PLASMA-LYTE A	potassium chloride	POTASSIUM phosphate	procainamide	proPOFol	REMIfentanil	ROcuronium	sodium bicarbonate	tobramycin	vancomycin	vasopressin
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vancomycin	C	C	C	C	I	I	I	I	C	C	C		C	C	C	✱	C	C	I	C	I			C	C		C	C	C	C	C	C	C	C	C	C	C	C	I	C		C	C	C			C	C	
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(Rev.27Aug2025)

Rapid Sequence Intubation (RSI)

RSI is an airway management technique that uses a sedative (induction agent) followed immediately by a neuromuscular blocking agent (NMBA) to create optimal conditions for rapid endotracheal intubation

It is the preferred method in securing a patient's airway in critically injured patients

Minimizes the risk of aspiration	Provides optimal intubation conditions by inducing unresponsiveness and muscle relaxation	Reduces physiological stress and complications

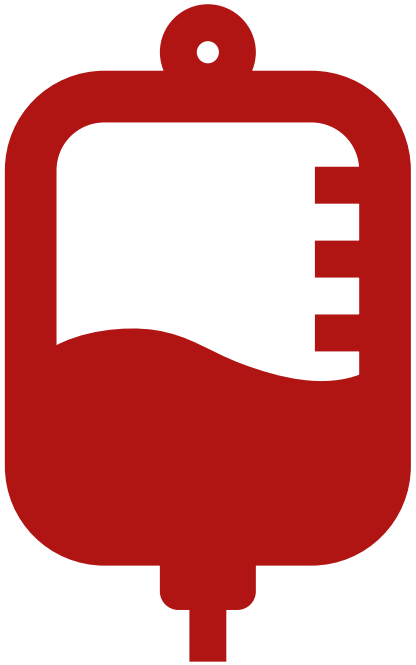
RSI Medications

NMBA's are agents that cause skeletal muscle paralysis by blocking transmission at the neuromuscular junction

The are used to create ideal intubating conditions by eliminating muscle tone and preventing spontaneous or airway reflexes.

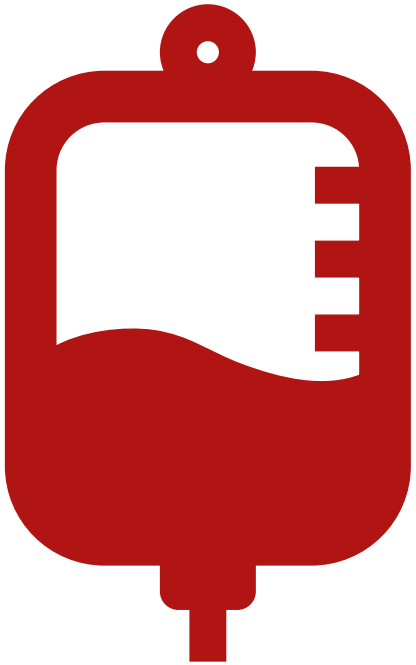
Two main categories	
Depolarizing NMBA	Nondepolarizing NMBA

Depolarizing NMBA



- **Succinylcholine 100 mg IV (1-1.5 mg/kg IV)**
- Works by depolarizing the motor endplate causing short acting paralysis
- Onset approx. 45-50 sec often with witnessed fasciulations
- Duration 5-9 min
- Does not have analgesic or sedative properties or NO reversal agent-Must wait it out!
- **Complications & Contraindications**
  - Hyperkalemia
  - Malignant Hyperthermia
  - Bradycardia
  - Prolonged paralysis-pseudeocholinesterase deficiency
  - Increased ocular pressure
  - Increased intragastric pressure & aspiration risk
  - Post operative myalgia –R/T the fasciculations preceding paralysis

Non-depolarizing NMBA



- **Rocuronium 1-1.5mg/kg**
- Works competitively blocking acetylcholine receptors, preventing depolarization and producing muscle relaxation
- Onset 1-2 min and duration 30-60 min
- Reversal agent: Sugammadex 2-4 mg/kg (preferred) or Neostigmine (slower to act on Rocuronium) 0.04-0.07 mg/kg up to max of 5 mg
- **Contraindications**
  - Known hypersensitivity to NMBA
  - Myasthenia Gravis (relative) may lead to prolonged paralysis
  - Use with caution
    - Chronic anticonvulsant therapy
    - Neuromuscular disease
    - Prolonged immobilization

Sedation

Propofol **2-2.5 mg/kg** for induction

- Infusion for ongoing sedation**: 5-50 mcg/kg/min or 0.3-3 mg/kg/hr or 50-300 mg/hr**
- Does not require dilution to make an infusion
- Requires vented tubing

Fentanyl **50-200 mcg** for induction

- Infusion for ongoing sedation/pain control: **12.5-200 mcg/hr**
- Mix as: 1000mcg into 100mL NS bag for 10 mcg/mL (can be also administered undiluted at 50 mcg/mL but need to consider IV rate required to 'push' med into patient)

Ketamine **1-2 mg/kg** for induction

- Infusion for ongoing sedation/pain control: **0.3-1.2 mg/kg/hr or 5-20 mcg/kg/min**
- Mix as: 100 mg in 100 mL NS bag for 1 mg/mL concentration (can be further concentrated to 2 mg/mL)

IV Fluid Administration

Normal Saline (NS, 0.9% NS)

- Isotonic IV fluid
- pH of 5.5
- More compatible with medications and blood

Lactated Ringers (LR, Ringers Lactate)

- Isotonic IV fluid
- Also contains sodium lactate, potassium, and calcium chloride
- pH of 6.5 (closer to that of blood at 7.4)
- Best for Burns
- Not compatible with blood administration
- Not compatible with many medications
- Giving LR DOES NOT increase lactate levels as it does not contain lactic acid

IV Fluid Administration

0.45% NS (half NS)

- Hypotonic IV solution
- Usually given with potassium or dextrose
- Depletes intravascular volumes and can cause cellular edema

3% NS (hypertonic saline)

- Hypertonic IV solution
- Utilized in severe head injury to reduce ICP or in seizures of unknown etiology
- Administer on a pump over 10 minutes or so in emergencies
- Corrects sodium rapidly and can lead to osmotic demyelination syndrome

IV Fluid Administration

Dextrose containing IV Fluids:

- Will lead to net fluid shifts out of cells
- Not used for resusciation
- D5W and D10W used to assist in slow correction of chronic hypernatremia or when hyponatremia has been corrected too rapidly

Potassium containing IV Fluids:

- Will increase osmolarity and cause fluids to become more hypertonic
- Adding potassium to a hypotonic base fluid such as D5 0.45%NS with potassium is an alternative for those with cellular dehydration and DKA instead of adding it to an isotonic fluid (as it may cause an isotonic fluid to become hypertonic)

Resources

Drug Monographs – the Ottawa, PEPID, 1 Cdn Fd Hosp

http://pdtm.vch.ca/monograph-information/monographs#group-p

Personal Cheat Sheets

EMCRit

LITFL

CrackCast

Y-site-Comp-Critical Care